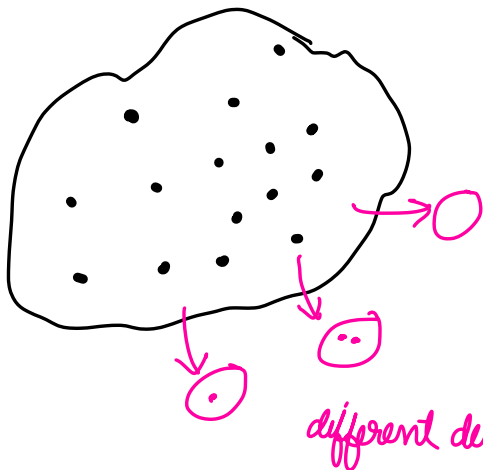



4/1/23

Advanced Heat Transfer

Forced convection $\rightarrow$ forcefully inject fluid to take away heat from the system

Natural convection $\rightarrow$ diff. in some property spatially to drive



If we enlarge our system, fluctuations will die out and will become a continuum

Open system $\rightarrow$ mass + energy both transfer

Closed $\rightarrow$ only energy transfer

standing x, y, z (fixed) $\leftarrow \frac{\partial}{\partial t}, \frac{d}{dt}, \frac{D}{Dt}$

$$\frac{dc}{dt} = \frac{\partial c}{\partial t} + \frac{\partial c}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial c}{\partial y} \cdot \frac{\partial y}{\partial t} + \frac{\partial c}{\partial z} \cdot \frac{\partial z}{\partial t}$$

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 v_x v_y v_z

$c(t, x, y, z)$

when going on a motorboat

$$\frac{Dc}{Dt} = \frac{\partial c}{\partial t} + \frac{\partial c}{\partial x} v_x + \frac{\partial c}{\partial y} v_y + \frac{\partial c}{\partial z} v_z$$

$\rightarrow$ here v_x, v_y, v_z are flow field velocity

$\hookrightarrow$ substantial derivative

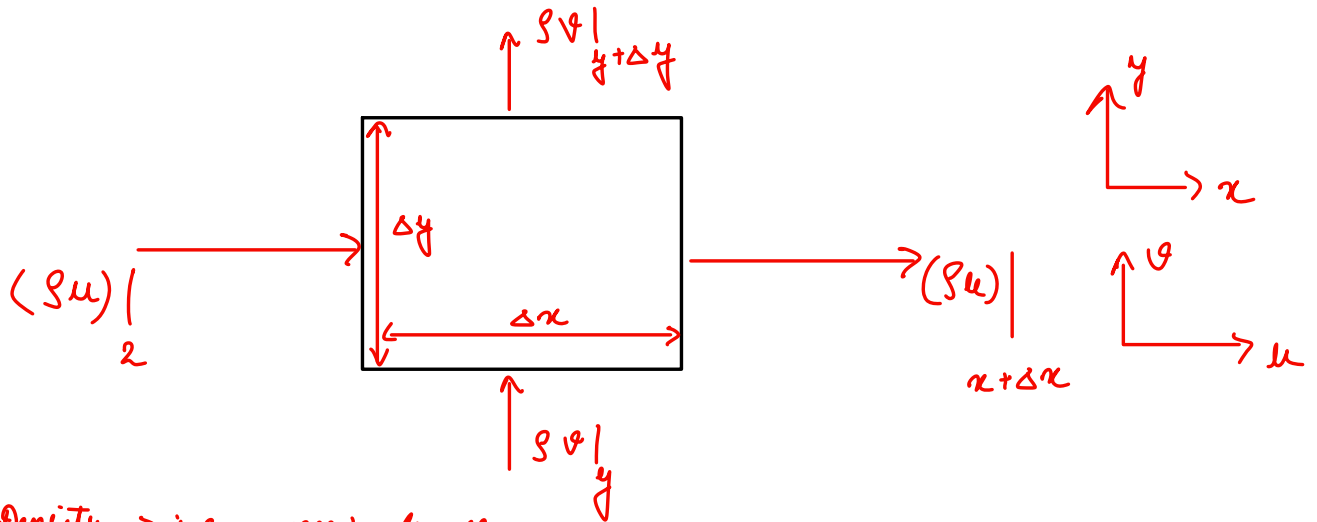
observe & property - that you are interested in, is being carried by same mass

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Transport operations

$$[\text{change in property in } \Delta t]_1 = \text{Rate of property [In-Out]}_2 + [\text{generation}]_3$$

mass transport



Density $\rightarrow$ i.e. mass/volume

$\hookrightarrow$ is getting transported here

$$\frac{\partial(\rho)}{\partial t} \times \Delta x \times \Delta y = \left[\cancel{\rho u \times \Delta y} + \cancel{\rho v \times \Delta x} \right] - \left[\cancel{\rho u} + \frac{\partial(\rho u)}{\partial x} \Delta x \right] \Delta y$$

$$\frac{\partial \rho}{\partial t} = - \left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} \right]$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0$$

$$\left[\cancel{\rho v} + \frac{\partial(\rho v)}{\partial y} \Delta y \right] \Delta x$$

$$\frac{\partial \rho}{\partial t} + \rho \frac{\partial u}{\partial x} + u \frac{\partial \rho}{\partial x} + \rho \frac{\partial v}{\partial y} + v \frac{\partial \rho}{\partial y} = 0$$

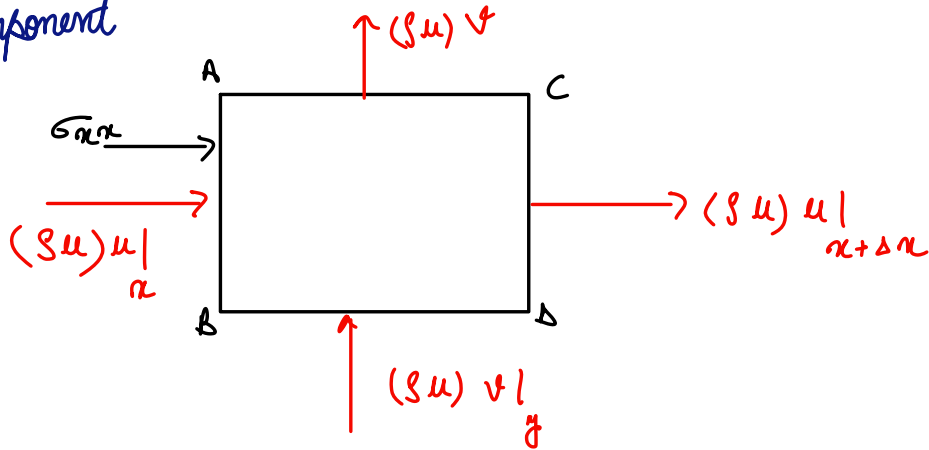
$$\frac{\partial \rho}{\partial t} + \rho \left[\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right] + \dots = 0$$

$$\frac{\partial \rho}{\partial t} + \rho (\nabla \cdot \vec{v}) = 0$$

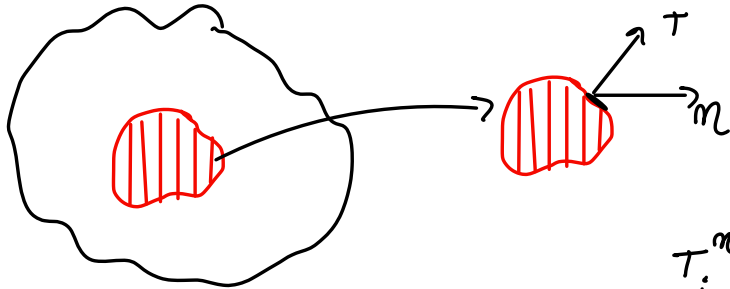
$\rho \rightarrow \text{constant} \rightarrow$ is not the necessary condition for incompressible fluid
 a specific condition
 $\frac{1}{\rho} \frac{\partial \rho}{\partial t} = 0 \rightarrow$ general condition
 (packet of fluid flowing through fluid is not getting compressed)

Momentum Transport

x -component



τ_{ij}



$$\tau_i^n \approx \tau_{ni} \approx \tau_{ji}$$

11/1/24

that function $\rightarrow$ only applicable for 2D flow

$$\rho C_p \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right]$$

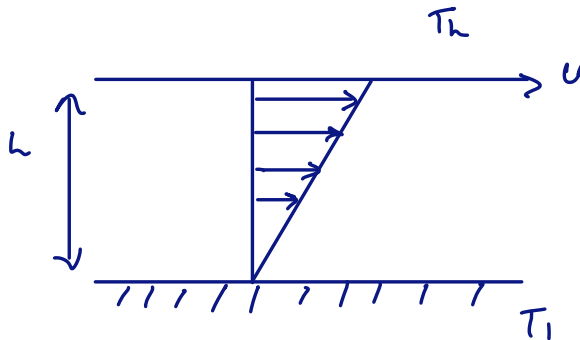
$$\frac{\partial}{\partial x} \left[\rho C_p u T - k \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[\rho C_p v T - k \frac{\partial T}{\partial y} \right] = 0$$

$$\frac{\partial H}{\partial y} = \rho C_p u (T - T_{ref}) - k \frac{\partial T}{\partial x}$$

$$-\frac{\partial H}{\partial x} = \underbrace{\rho C_p v (T - T_{ref})}_{\text{convective part}} - \underbrace{k \frac{\partial T}{\partial y}}_{\text{diffusive part}}$$

$H \rightarrow$ fn. of both v, T
 $H(v, T)$

①



$$u = U \left(\frac{y}{L} \right)$$

$$T_2 > T_1$$

lubricating oil $\rightarrow$ viscous effect imp

assumptions 1) SS flow

$$2) \dot{q}''' = 0$$

3) Incompressible $\beta = 0$

$$0 = k \frac{d^2 T}{dy^2} - \frac{d}{dy} (\tau_{yx} \times u)$$

$\downarrow$
 no need for partial, as 1D

$$\tau_{yx} = -\mu \frac{\partial u}{\partial y}$$

$\checkmark$
 substituting & expanding

viscous dissipation
 general expression

$$\downarrow$$

$$-\nabla \cdot (\tau \cdot \vec{v})$$

Transient + Bulk motion

substantial
 derivative

$$\left(\mu \mu \frac{d^2 u}{dy^2} + \mu \left(\frac{du}{dy} \right)^2 \right)$$

$$0 = \frac{k d^2 T}{dy^2} + \left[\mu \left(\frac{u}{L} \right)^2 \right] \quad \text{--- ①}$$

$$\theta = \frac{T(y) - T_1}{T_L - T_1} \quad \eta = y/L$$

$$\frac{d\theta}{d\eta} = \frac{d\theta}{dy} \times \frac{dy}{d\eta}$$

$$= \frac{1}{(T_L - T_1)} \times \frac{dT}{dy} \times L$$

$$\frac{d^2 \theta}{d\eta^2} = \frac{L^2}{(T_L - T_1)} \frac{d^2 T}{dy^2}$$

$$0 = \frac{k (T_L - T_1)}{L^2} \frac{d^2 \theta}{d\eta^2} + \mu \left(\frac{u}{L} \right)^2$$

$$\frac{d^2 \theta}{d\eta^2} = - \frac{\mu u^2}{k (T_L - T_1)} = -Br \quad (\text{Brinkman number})$$

$$\theta(\eta) = \frac{-Br \eta^2}{2} + c_1 \eta + c_2$$

Boundary conditions

$$y=0 \quad T=T_1$$

$$y=L \quad T=T_L$$

$$\eta=1 \quad \theta=1$$

$$\eta=0 \quad \theta=0$$

$$c_1 = 1 + \frac{Br}{2}$$

$$\theta(\eta) = \frac{-Br}{2} \eta^2 + \left(1 + \frac{Br}{2} \right) \eta$$

$$\theta(\eta) = \frac{Br}{2} (\eta - \eta^2) + \eta$$

(ii) Heat flux at moving plate

$$q = -k \frac{dT}{dy}$$

$$= -K \frac{(T_h - T_l)}{L} \frac{d\theta}{dm}$$

$$q|_{y=L} = -K \frac{(T_h - T_l)}{L} \left[1 - \frac{B_{Tc}}{2} \right]$$

$$\left[\frac{d\theta}{dm} = 1 + (1 - 2m) \frac{B_{Tc}}{2} \right]$$

$$\text{If } B_{Tc} = 2, q|_{y=L} = 0$$

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$Br =$ Brinkman number

$$= \frac{\mu v^2}{k(\Delta T)} \rightarrow \text{viscous} \\ \rightarrow \text{conductive}$$

$$= \frac{U \times \mu \times \frac{v}{\Delta y}}{k\left(\frac{\Delta T}{\Delta y}\right)} \rightarrow \text{viscous generation} \\ \rightarrow \text{this is conduction of that heat generated}$$

high Brinkman number little mismatch b/w generation & conduction

$$\rho C_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial y^2}$$

scaling analysis

1. $c = a + b$

$$O(c) \sim O(a) \sim O(b)$$

$\hookrightarrow$ represents similar order

$$O(a) \gg O(b)$$

$$O(c) \sim O(a)$$

2. $c = a \times b$

$$O(c) \sim O(a) \times O(b)$$

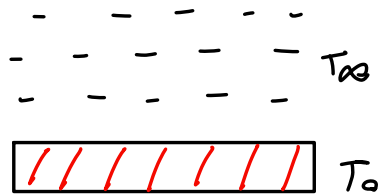
$$F = \int_A \tau_0 dA$$

$$\tau_0 = -\mu \left(\frac{\partial u}{\partial y} \right)$$

$$u = f(y)$$

Navier Stokes assumptions

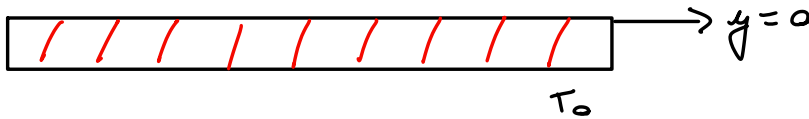
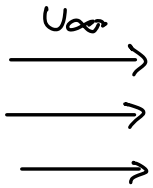
1. Newtonian
2. Incompressible



heat flow rate $\leftarrow \dot{q} = \int_A q'' dA \rightarrow$ heat flux

$$q'' = h(\Delta T) \quad [\text{Newton's Law}]$$

how to estimate this?



$$q'' = -k \left(\frac{\partial T}{\partial y} \right)_{y=0}$$

$\rightarrow$ only conduction, as relative vel 0 at $y=0$ [No slip]

$$-k \left(\frac{\partial T}{\partial y} \right)_{y=0} = h \Delta T$$

$\rightarrow$ At steady state

$$h = \frac{-k \left(\frac{\partial T}{\partial y} \right)_{y=0}}{(\Delta T)}$$

If you estimate h , you estimate q'' , and hence $\dot{q}$

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scale

$$y \sim \delta$$

$$x \sim L$$

$$u \sim V_\alpha$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\sim U_\alpha \frac{U_\alpha}{L} \sim \nu \frac{U_\alpha}{\delta} \sim \frac{\rho_\alpha}{\rho L} \sim \cancel{\nu \frac{U_\alpha}{L^2}} \sim \nu \frac{U_\alpha}{\delta^2}$$

so neglected by order of magnitude analysis

Assumption $\delta \ll L$

Now checking scale of continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{U_\alpha}{L} \sim \frac{V}{\delta} \Rightarrow V \sim U_\alpha \left(\frac{\delta}{L} \right) \rightarrow \text{substitute this}$$

$$\sim \frac{U_\alpha^2}{L} \sim \frac{U_\alpha^2}{L}$$

same order can't cancel/neglect

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \frac{\partial^2 u}{\partial y^2} \right)$$

same order, so did not cancel

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$P = P(x, y)$$

$$dP = \frac{\partial P}{\partial x} dx + \frac{\partial P}{\partial y} dy$$

$$\frac{dP}{dx} = \frac{\partial P}{\partial x} + \frac{\partial P}{\partial y} \frac{dy}{dx}$$

$$\frac{dP/dx}{\partial P/\partial x} = 1 + \left(\frac{\frac{\partial P}{\partial y} \frac{dy}{dx}}{\frac{\partial P}{\partial x}} \right) \quad - (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial P}{\partial x} + \nu \left(\cancel{\frac{\partial^2 u}{\partial x^2}} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\frac{1}{\rho} \frac{\partial P}{\partial x} \sim \frac{\nu U_\infty}{\delta^2}$$

→ pressure force balanced by viscous force

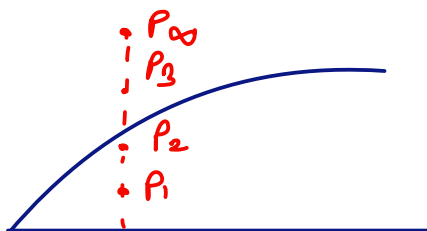
$$\frac{1}{\rho} \frac{\partial P}{\partial x} \sim \frac{\nu \rho}{\delta^2} \quad - (2)$$

From (1) & (2)

$$\frac{\frac{\nu \rho}{\delta^2} \times \frac{\delta}{h}}{\frac{\nu U_\infty}{\delta^2}} \sim \frac{\nu}{U_\infty} \left(\frac{\delta}{h} \right)$$

$$\frac{\delta}{h} \ll 1$$

$$\text{So, } \frac{dP}{dx} \sim \frac{\partial P}{\partial x} \sim \frac{dP_x}{dx}$$



→ P varying only with x, not y, so $P_1 = P_2 = P_3 = P_0$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \left(\frac{\partial^2 u}{\partial y^2} \right)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$u, v, T \rightarrow \text{variables}$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

our aim was to get τ, q''

$$\tau_{\infty} \sim \mu \frac{U_{\infty}}{\delta} \quad q'' \sim h(\Delta T)$$

$$h \sim \frac{k}{\delta_T} \rightarrow \text{from previous class}$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \left(\frac{\partial^2 u}{\partial y^2} \right)$$

$$\sim \frac{U_{\infty}^2}{L}$$

$$\sim \frac{\nu U_{\infty}}{\delta^2}$$

At SS conduction = convection

$\sim \text{conduction} \sim \text{diffusion}$

$$\frac{U_{\infty}^2}{L} \sim \frac{\nu U_{\infty}}{\delta^2}$$

$$\delta^2 \sim \frac{\nu L}{U_{\infty}}$$

$$\frac{\delta^2}{L^2} \sim \left(\frac{\nu}{U_{\infty} L} \right)$$

$$\frac{\delta}{L} \sim (Re_L)^{-1/2}$$

$$\tau \sim \mu \frac{U_{\infty}}{\delta}$$

$$\sim \mu \frac{U_{\infty}}{LX(Re)^{-1/2}}$$

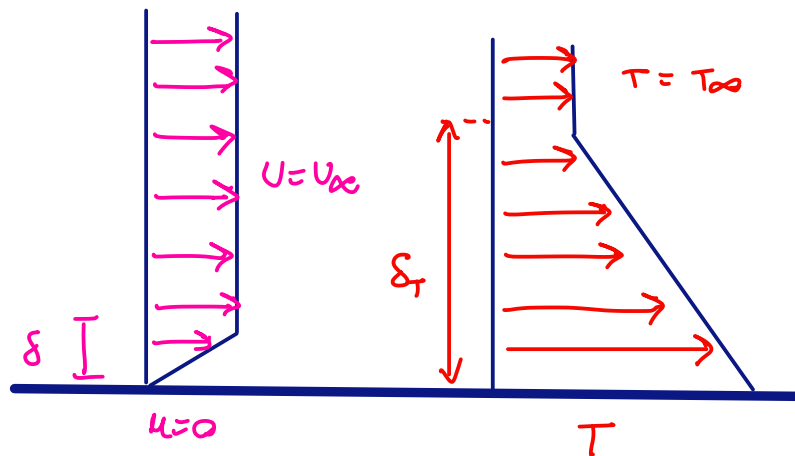
$$\tau \sim \rho U_{\infty}^2 Re_h^{-1/2}$$

$$C_f \sim \frac{\tau}{\rho U_{\infty}^2} \sim Re_h^{-1/2}$$

here we have assumed properties are decoupled (fluid & heat transfer)

↓
 μ, ρ change with temp. is not considered

when $\delta \ll \delta_T$



At the edge of Thermal Boundary Layer

$$U \sim U_{\infty}$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{U_{\infty}}{L} \sim \frac{v}{\delta_T}$$

$\Rightarrow$

$$v \sim U_{\infty} \left(\frac{\delta_T}{L} \right)$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial y^2}$$

$$U_{\infty} \frac{\Delta T}{L} \sim \frac{v \Delta T}{\delta_T} \sim \alpha \frac{\Delta T}{\delta_T^2}$$

$$u_{\infty} \frac{\Delta T}{h} \sim \alpha \frac{\Delta T}{\delta_T^2}$$

At SS, convection $\sim$ diffusion

$$\delta_T^2 \sim \frac{\alpha h}{u_{\infty}}$$

$$\frac{\delta_T^2}{h^2} \sim \left(\frac{\alpha}{u_{\infty} h} \right) \rightarrow \text{Peclet Number}$$

$$\frac{\delta_T}{h} \sim Pe^{-1/2} Re^{-1/2}$$

$$Pe = Re \times Pr$$

$$h \sim \frac{k}{\delta_T} \sim \frac{k}{h} \times Re^{1/2} \times Pr^{1/2}$$

$$Nu \sim Re^{1/2} Pr^{1/2}$$

$$\frac{\delta}{\delta_T} \sim Pr^{1/2} \ll 1 \rightarrow \text{so we started with a condition } \frac{\delta}{\delta_T} \ll 1$$

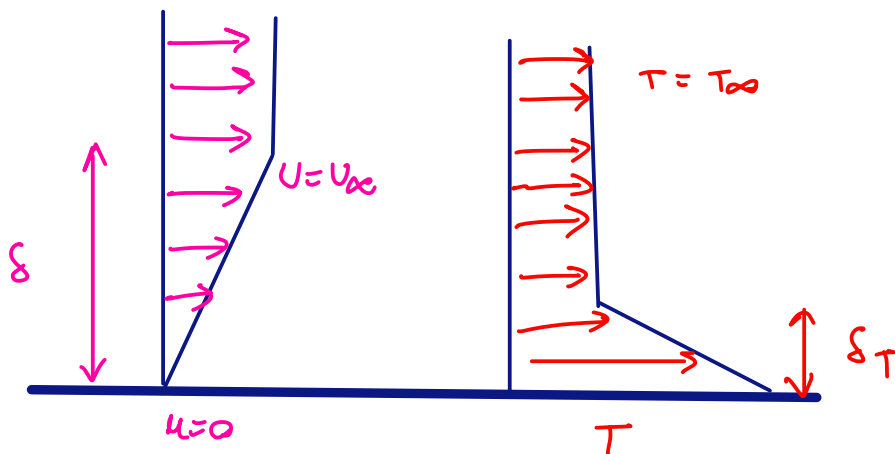
and got to know that it is valid
only for liquid metals

(Taught by SDG sir also!!!)

Liquid metals be case in $\delta \ll \delta_T$!!!

When

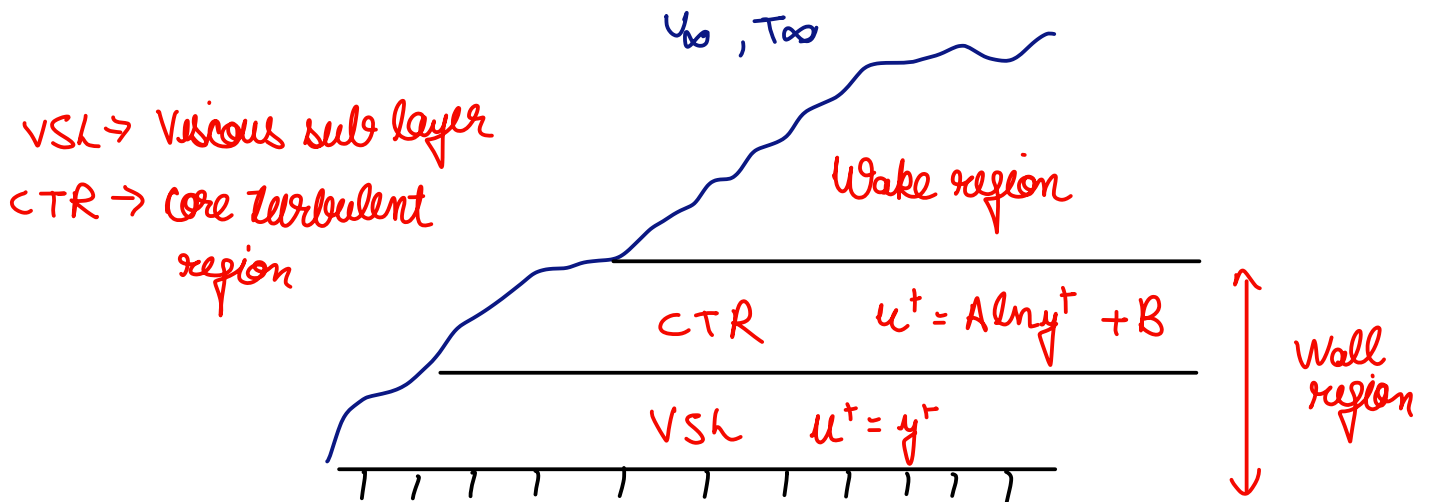
$$\delta \gg \delta_T$$



At the edge of Thermal Boundary Layer

$$u \sim u_{\infty} \left(\frac{\delta_T}{\delta} \right)$$

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Turbulent boundary layer

Three velocity profiles, for three different regions

$$u^+ = \frac{\bar{u}}{u^+} \quad u^+ = \left(\frac{T_0}{S}\right)^{1/2} \quad \text{we don't know } T_0$$

Wake region no expression unknown, so some people extrapolated in wake region

$$u^+ = f_n(y^+)$$

$$\frac{\bar{u}}{u^+} = f_n\left(\bar{y}\left(\frac{u^+}{v}\right)\right)$$

$\rightarrow$ at edge of hydrodynamic BL, $\bar{y} = \delta$

$$\left(\frac{u_\infty}{T_0/S}\right)^{1/2} = f_n\left(\delta\left(\frac{T_0/S}{v}\right)^{1/2}\right) \quad \text{--- (1)} \quad T_0, \delta \rightarrow \text{unknown}$$

$$\frac{d}{dx} \int_0^\delta u(u_\infty - u) dy = \frac{1}{S} \cancel{\gamma \frac{dp_0}{dx}} + \cancel{\frac{dv_0}{dx}} \int_0^\delta u dy + v \left(\frac{\partial u}{\partial y}\right)_{y=0} \sim \left(\frac{T_0}{S}\right)$$

$\rightarrow$ fn of δ

$$\frac{d}{dx} \int_0^\delta u(u_\infty - u) dy = \left(\frac{T_0}{S}\right) \quad \text{--- (2)} \quad 2 \text{ eqn, 2 unknown}$$

$$u^+ = 8.7 (y^+)^{1/7}$$

→ written in wall coordinates

↳ Prandtl 1/7th power law

$T_o \rightarrow$ was assumed to be constant

→ There was no inertia

$q_o'' \rightarrow$ will be assumed to be constant

→ No effect of inertia

→ that flux from wall
is assumed

$$\frac{\partial}{\partial y} \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right] = 0$$

$$(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} = \text{constant} = \frac{-q_o''}{Sc_p}$$

$$y^+ = \bar{y} \left(\frac{u^+}{\nu} \right)$$

$$\left(\frac{\nu}{u^+} \right) dy^+ = d\bar{y}$$

$$\frac{Sc_p}{q_o''} = \partial(T_o - \bar{T})$$

$$= \frac{d\bar{y}}{(\alpha + \epsilon_H)}$$

$$\frac{Sc_p u^+}{q_o''} \partial(T_o - \bar{T}) = \frac{dy^+}{(\alpha/\nu + \epsilon_H/\nu)}$$

$$\frac{Sc_p u^+}{q_o''} d(T_o - \bar{T}) = \frac{dy^+}{\left(\frac{1}{Pr_t} + \left(\frac{\epsilon_H}{\epsilon_M} \right) \times \left(\frac{\epsilon_M}{\nu} \right) \right)}$$

Non dimensional temp

$1/Pr_t \rightarrow$ Turbulent
Prandtl Number

$$\tau_{app} = \mu \frac{\partial \bar{u}}{\partial y} - \rho \overline{u'v'} = \rho(\nu + \epsilon_M) \frac{\partial \bar{u}}{\partial y} \quad \text{apparent shear stress}$$

$$-q_{app}'' = k \frac{\partial \bar{T}}{\partial y} - \rho c_p \overline{v'T'} = \rho c_p (\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \quad \text{apparent heat flux}$$

No inertia

$$\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = -\frac{1}{\rho} \frac{d\bar{P}}{dx} + \frac{\partial}{\partial y} \left[(\nu + \epsilon_M) \frac{\partial \bar{u}}{\partial y} \right]$$

$$\bar{u} \frac{\partial \bar{T}}{\partial x} + \bar{v} \frac{\partial \bar{T}}{\partial y} = \frac{\partial}{\partial y} \left[(\alpha + \epsilon_H) \frac{\partial \bar{T}}{\partial y} \right]$$

↳ No inertia

$$T^+(x^+, y^+) = \frac{\rho C_p u^*}{q_w''} (T_0 - \bar{T}) \rightarrow \text{Non dimensionalised Temp (Defined)}$$

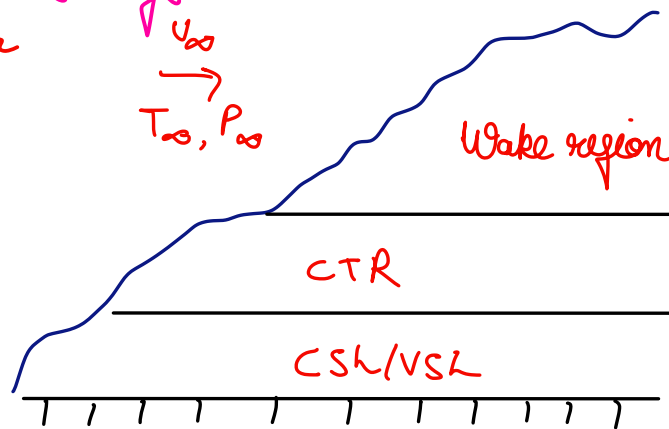
$$dT^+ = \frac{dy^+}{\left(\frac{1}{Pr_t} + \frac{1}{Pr_t} \left(\frac{\epsilon_M}{\nu} \right) \right)}$$

$$u^* = \left(\frac{T_0}{\rho} \right)^{1/2}$$

This term represents eddies

u_∞, T_∞

CSL $\rightarrow$ conduction sub layer



$$dT^+ = \int_0^{y_{CSL}^+} \frac{dy^+}{\left(\frac{1}{Pr_t} + (\text{neg term}) \right)} + \int_{y_{CSL}^+}^{y^+} \frac{dy^+}{\left((\text{neg term}) + \frac{1}{Pr_t} \left(\frac{\epsilon_M}{\nu} \right) \right)}$$

$0 < y^+ < y_{CSL}^+$ in CSL region

$$\int_{y_{CSL}^+}^{y^+} dT^+ = \int_{y_{CSL}^+}^{y^+} \frac{dy^+}{\left((\text{neg term}) + \frac{1}{Pr_t} \left(\frac{\epsilon_M}{\nu} \right) \right)}$$

$$T^+ - T_{CSL}^+ = Pr_t \int \frac{dy^+}{\left(\frac{\epsilon_M}{\nu} \right)} \quad -a$$

$$u^+ = \frac{1}{K} \ln y^+ + B$$

$$\frac{du^+}{dy^+} = \frac{1}{K y^+}$$

$$\frac{\varepsilon_M}{\nu} \frac{du^+}{dy^+} = 1$$

$$\frac{\varepsilon_M}{\nu} = \frac{dy^+}{du^+} = K y^+ \quad \text{--- (1)}$$

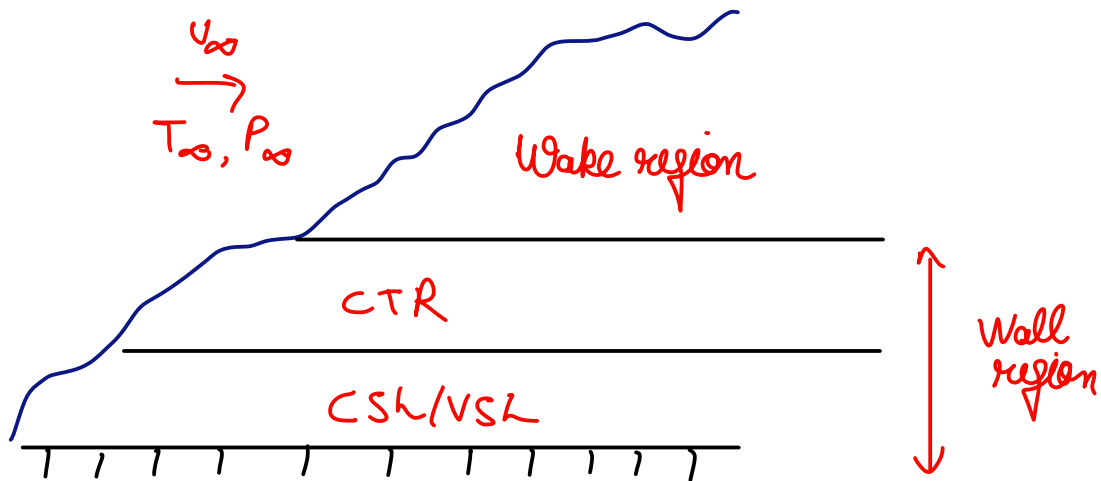
Here we are assuming that ν_{SL} & ε_{SL} are same, which may not be the case

Solving (1) by putting (1)

$$T^+ = Pr y_{CSL}^+ + \frac{Pr}{K} \ln \frac{y^+}{y_{CSL}^+} \quad \text{--- (2)} \rightarrow \text{In CTR region}$$

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Turbulent BL



Velocity distribution $\rightarrow$ Assuming that $T_0 \neq f(y)$

$$u^+ = y^+ \quad 0 < y^+ < y_{VSL}^+$$

$$u^+ = \frac{1}{K} \ln y^+ + B \quad y_{VSL}^+ < y^+ < y^+$$

$$K = 0.4$$

$$B = 5.5$$

Temperature distribution $\rightarrow$ assuming that $q_0'' \neq f(y)$

$$T^+ = P_{\infty} y^+ \quad 0 < y^+ < y_{CSL}^+$$

$$T^+ = P_{\infty} y_{CSL}^+ + \frac{P_{\infty t}}{K} \ln \frac{y^+}{y_{CSL}^+}$$

$$y_{CSL}^+ \approx 13.2 \quad P_{\infty t} \approx 0.9 \quad K \approx 0.41 \quad 0 \leq P_{\infty} \leq 5$$

$\hookrightarrow$ Experimental values

$$T^+ \approx 2.195 \ln y^+ + 13.2 P_{\infty} - 5.66 \rightarrow \text{after plugging in the values}$$

$$T^+(x^+, y^+) = \frac{\rho C_p u^*}{q_0''} (T_0 - T) \quad u^* = \left(\frac{\tau_0}{\rho} \right)^{1/2}$$

$q_0'', T_0 \rightarrow \text{both unknown}$

$$\frac{\rho C_p u^* (T_0 - T)}{q_0''} = P_{\infty} y_{CSL}^+ + \frac{P_{\infty t}}{K} \ln \left(\delta \left(\frac{u^*}{v} \right) \right) - \frac{P_{\infty}}{K} \ln y_{CSL}^+ \quad (1)$$

$\hookrightarrow$ Trying to extrapolate CTR in wake region $\checkmark \rightarrow \text{unknowns}$
 since P_{∞} is of order 1, $\delta = \delta_T$, so used δ instead of δ_T

$$u^+ = \frac{\bar{u}}{u^*} = \frac{1}{K} \ln y^+ + B \quad \rightarrow \text{at edge of BL}$$

$$u^+ = \frac{u_{\infty}}{u^*} = \frac{1}{K} \ln \left(\delta \frac{u^*}{v} \right) + B \quad (2)$$

$$\frac{u_{\infty}}{u^*} = \left(\frac{2}{C_{Fx}} \right)^{1/2} \rightarrow \text{condition imposed}$$

$$\left(\frac{u_{\infty}}{u^*} \right)^2 = \frac{u_{\infty}^2}{\tau_0 / \rho} = \frac{2}{C_{Fx}}$$

$$C_{Fx} = \frac{\tau_0}{\frac{1}{2} \rho u_{\infty}^2}$$

$$\tau_0 = \frac{1}{2} \rho u_{\infty}^2 C_{Fx}$$

$$\frac{h}{8 C_p U_\infty} = \frac{C_f x / 2}{Pr_t + \left(\frac{1}{2} C_f x\right) \left[Pr_t y_{cs, h}^+ - B Pr_t - \left(\frac{Pr_t}{K}\right) \ln y_{cs, h}^+ \right]}$$

→ St_x

↓
as h is fn. of x , so x subscript
↪ after solving ① & ②

$$St_x = \frac{1/2 C_f x}{0.9 + \left(\frac{1}{2} C_f x\right)^{1/2} (13.2 Pr_t - 10.25)}$$

→ after putting values

$C_f x$ → weak fn. of Re number

$$St_x \propto \frac{1}{2} C_f x$$

$$= K \frac{1}{2} C_f x \rightarrow \text{fn.}(Pr_t)$$

$$St_x Pr_t^{2/3} = \frac{1}{2} C_f x$$

→ Colburn relation

Based on Prandtl 2 layer theory

$$St_x = \frac{1/2 C_f x}{1 + 5 \left(\frac{1}{2} C_f x\right)^{1/2} \left[Pr_t - 1 + \ln \left[1 + \frac{5}{6} (Pr_t - 1) \right] \right]}$$

↪ Based on 3 layer theory

Put $Pr_t = 1$, we get $St_x = \frac{1}{2} C_f x$

$$\frac{T_0}{s} = (\alpha + \epsilon_m) \frac{d\bar{u}}{dy}$$

$$T_0 = (\mu + s \epsilon_m) \frac{d\bar{u}}{dy} \quad \text{--- ①}$$

$$-\frac{q_{o''}}{8 C_p} = (\alpha + \epsilon_h) \frac{d\bar{T}}{dy}$$

$$-q_0'' = (k + s c_p \epsilon_H) \frac{dT}{dy} \quad - (2)$$

$$(1) \div (2)$$

$$\frac{T_0}{-q_0''} = \frac{(\mu + s \epsilon_M)}{(k + s c_p \epsilon_H)} \frac{d\bar{u}}{dT}$$

$$Pr = \frac{\mu c_p}{k}, \text{ if } Pr_t = 1 \quad \epsilon_H = \epsilon_M \text{ \& } Pr = 1, \text{ we get}$$

$$\frac{T_0}{q_0''} = \frac{1}{c_p} \frac{d\bar{u}}{dT}$$

→ not exactly, but order 1

$$\frac{T_0}{q_0''} = \frac{1}{c_p} \frac{u_\infty}{(T_\infty - T_0)} \rightarrow \text{at edge}$$

$$\frac{1/2 s \cancel{u_\infty^2} c_f x}{h \cancel{\Delta T}} = \frac{1}{c_p} \frac{\cancel{u_\infty}}{\cancel{\Delta T}}$$

$$St_x = \frac{h}{u_\infty s c_p} = \frac{1}{2} c_f x$$

↙
Reynold's analogy

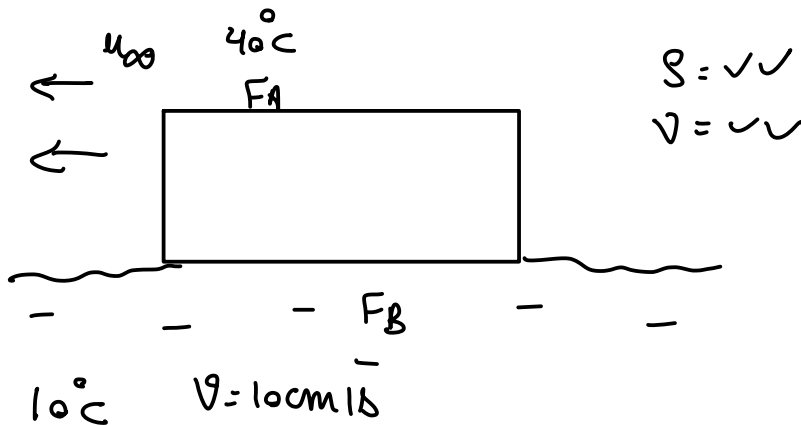
→ ϵ_M / ϵ_H
 $Pr_t = 1$ physically means that instead of car, we have bigger tanks carrying same capacity of water & gas cylinder. If molecules have same capacity of carrying energy and momentum then $Pr_t = 1$.

c_x car having same capacity to carry gas cylinder & water cylinder

Add 5/3
Tree Notes

Problem 4

A flat sheet (tabular) iceberg drifts over the ocean as it is driven by the wind that blows over the top. The temperature of the surrounding seawater is 10°C , and the relative velocity between it and the iceberg is 10 cm/s . The length of the iceberg in the direction of drift is L . Calculate the corresponding wind velocity when the atmospheric air temperature is 40°C .



$$F_A = F_B \text{ as } 0 \text{ acceleration}$$

$$\tau_A \times A = \tau_B \times A$$

$$\tau_A = \tau_B$$

$$\tau_A = 0.037 \times \rho_a \times u_a^2 \times \left(\frac{u_\infty L}{\nu_a} \right)^{-1/5}$$

$$\tau_B = 0.03 \times \rho_w \times u_w^2 \times \left(\frac{u_w \times h}{\nu_B} \right)^{-1/5}$$

only u_a is unknown

Ans $\Rightarrow u_a = 11.7 \text{ km/h}$

Problem 5

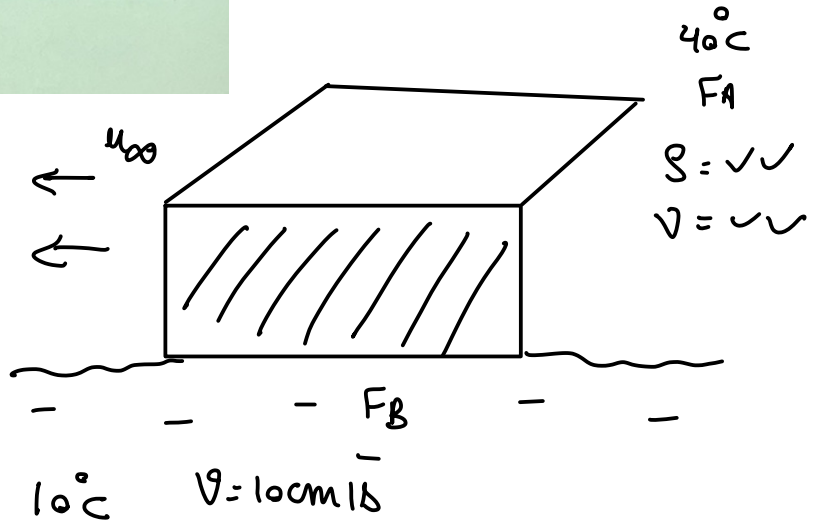
Air flows with velocity 3.24 m/s over the top surface of the flat iceberg discussed in Problem 7.14. The air temperature outside the boundary layer is 40°C , and the ice surface temperature is 0°C . The length of the iceberg in the direction of air flow is $L = 100$ m. The ice latent heat of melting is $h_f = 333.4$ kJ/kg.

Calculate the L -averaged heat flux deposited by the air flow into the upper surface of the iceberg (model this surface as flat). Calculate, in millimeters per hour, the rate of melting caused by this heat flux, that is, the erosion (thinning) of the ice slab.

→ similar to above

→ Need to know the surface heat flux → $h\Delta T$

→ unless stated in ques don't use correlations



$$q'' = h(\Delta T)$$

$$Nu = \frac{hL}{k} \rightarrow h = \frac{Nu \times k}{L}$$

$$\dot{m} = 2.66 \text{ m/hr}$$

Ch 7 Bejan Book → exercise problems

Natural convection

Volumetric strain rate $\frac{DV}{Dt} = 0 \rightarrow$ necessary condition for incompressible flow → $S = \text{constant}$ (special case)

$$\frac{1}{V} \frac{DV}{Dt} = \nabla \cdot \vec{V}$$

→ Volumetric strain rate per unit volume = gradient of velocity (can be proved)

$$m = S \times V$$

$$\ln m = \ln S + \ln V$$

$$\frac{1}{m} \frac{Dm}{Dt} = \frac{1}{S} \frac{DS}{Dt} + \frac{1}{V} \frac{DV}{Dt}$$

conservation of mass

$$\text{so, } \boxed{\frac{DS}{Dt} = 0}$$

→ = 0 for incompressible fluid

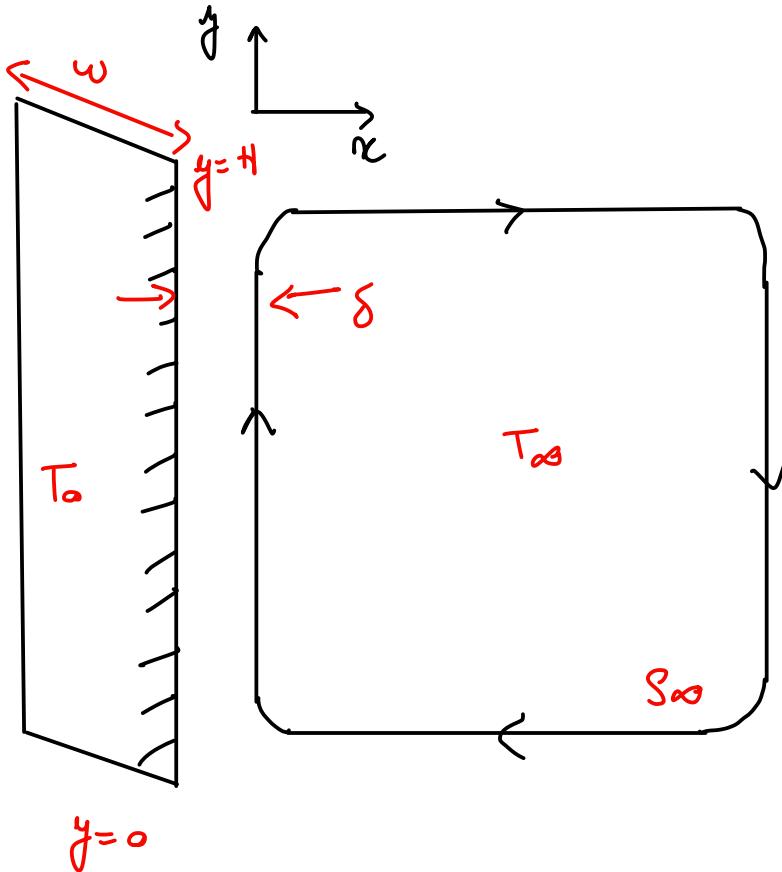
$$\frac{\partial S}{\partial t} + u \frac{\partial S}{\partial x} + v \frac{\partial S}{\partial y} = 0 \rightarrow \text{here } S \text{ may or may not be constant}$$

→ so natural convection → density may vary, yet incompressible flow

$$\frac{DS}{Dt} = 0, \text{ does not}$$

$$\frac{DV}{Dt} = 0$$

mean S is constant, may vary



$$T_0 > T_\infty$$

$$H \gg \delta$$

$$Q = (w \times H) \times \underbrace{h_{o-H}}_{\text{avg heat transfer coefficient}} \times (T_0 - T_\infty)$$

Assumption: 2D flow

Incompressible flow

Here velocity & Temp are coupled → as S changing → driving the flow

↓
v & T coupled

so, need to go to basic eqns

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$S \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = -\frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$S \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = -\frac{\partial P}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - Sg \rightarrow \text{-ve } g \text{ downward}$$

x dimension $\ll y$ dimension

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right]$$

o again by dimension

equations finally remaining

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad -①$$

from BL, just that here it is vertical

instead of $-\frac{dP_\infty}{dx}$

we use $-\frac{dP_\infty}{dy}$

$$s \left[u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right] = -\frac{dP_\infty}{dy} + u \frac{\partial^2 v}{\partial x^2} - sg$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = \alpha \frac{\partial^2 T}{\partial x^2} \quad -③$$

$$\frac{dP_\infty}{dy} = -s_\infty g \rightarrow \text{hydrostatic pressure}$$

$$s \left[u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right] = u \frac{\partial^2 v}{\partial x^2} + g(s_\infty - s) \quad -②$$